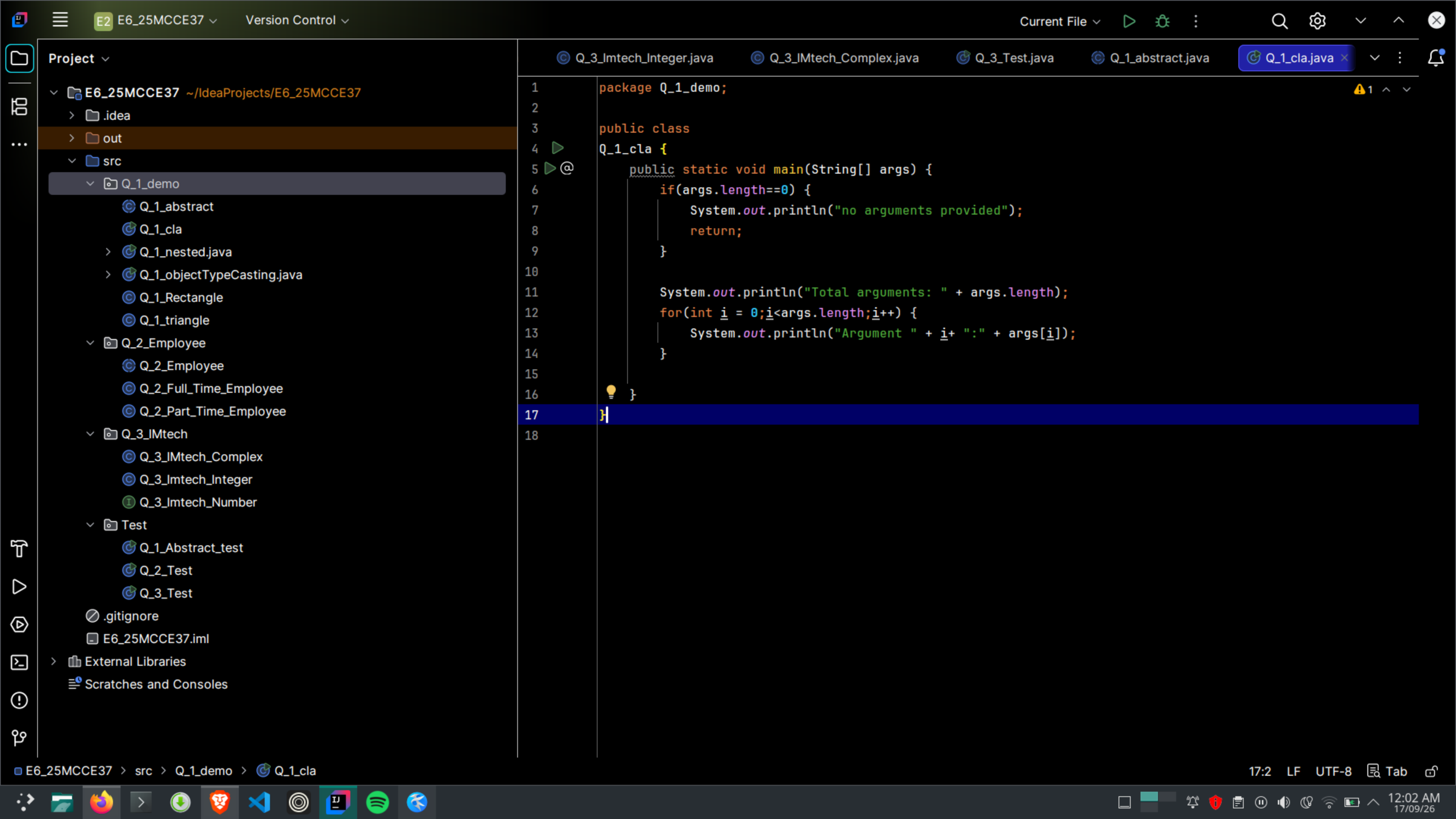


The image shows a screenshot of an IDE interface. On the left is a project explorer showing a project named 'E6_25MCCE37' with a directory structure including 'src' and 'Q_1_demo'. The file 'Q_1_abstract' is selected. The main editor displays the code for 'Q_1_abstract.java', which is a public abstract class with two protected double attributes, a constructor, and an abstract method 'area()'. The code is as follows:

```
1 package Q_1_demo;
2
3 public abstract class Q_1_abstract { 3 usages 2 inheritors
4     protected double dim1; 3 usages
5     protected double dim2; 3 usages
6     public Q_1_abstract(double dim1, double dim2) { 2 usages
7         this.dim1 = dim1;
8         this.dim2 = dim2;
9     }
10    public abstract double area(); 1 usage 2 implementations
11 }
```

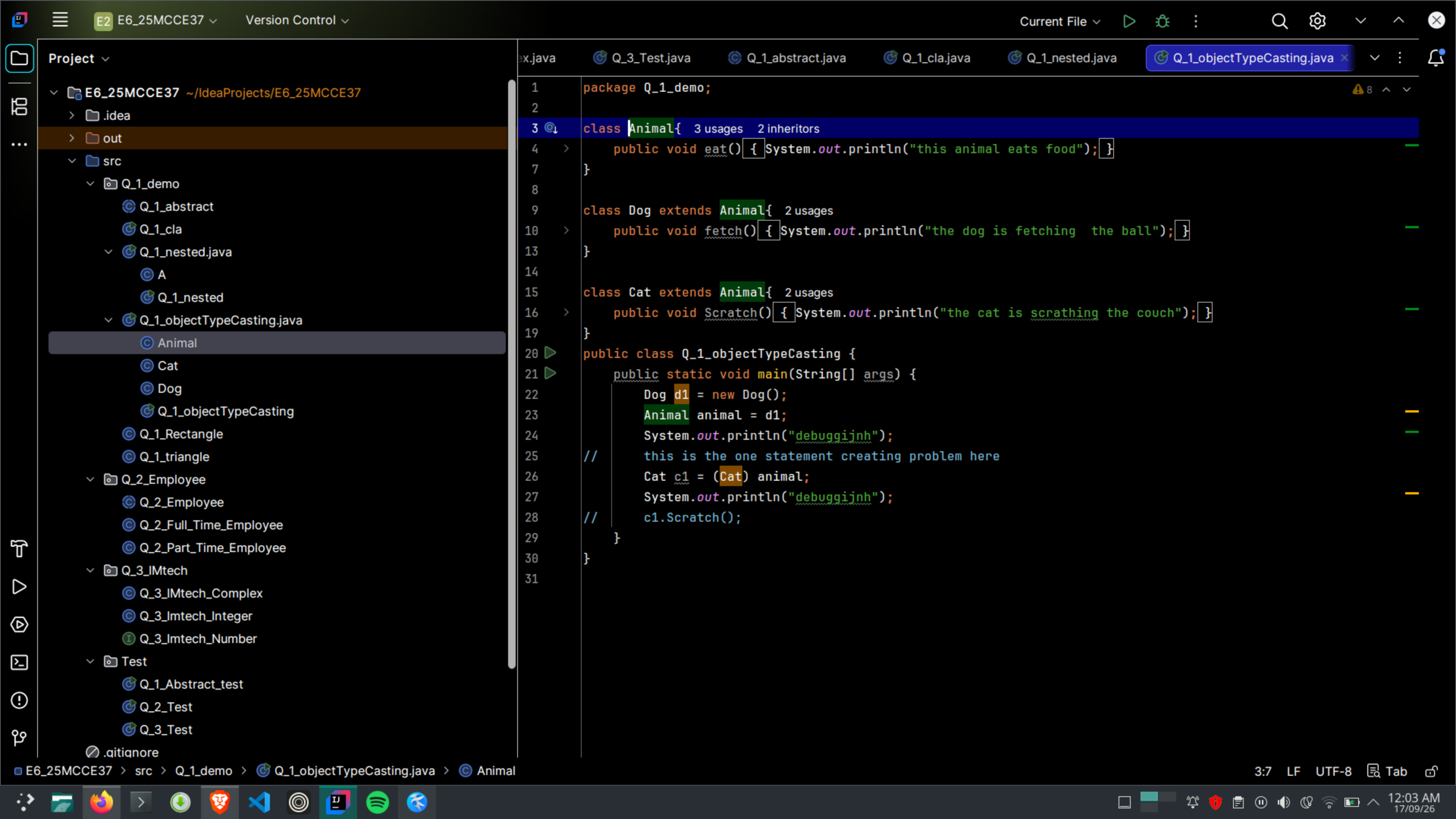
The bottom status bar shows the current file path 'E6_25MCCE37 > src > Q_1_demo > Q_1_abstract' and the time '12:01 AM 17/09/26'.

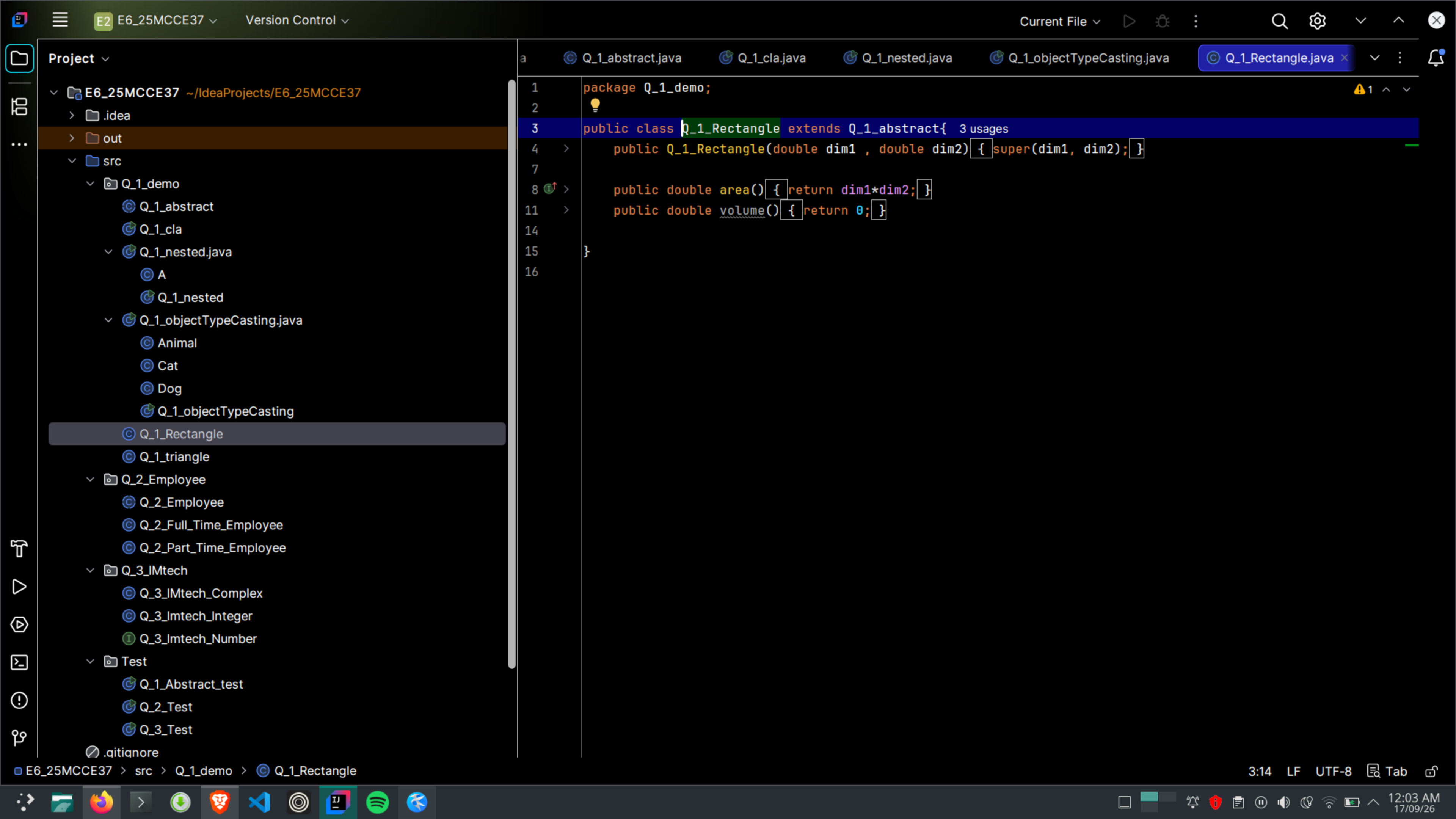


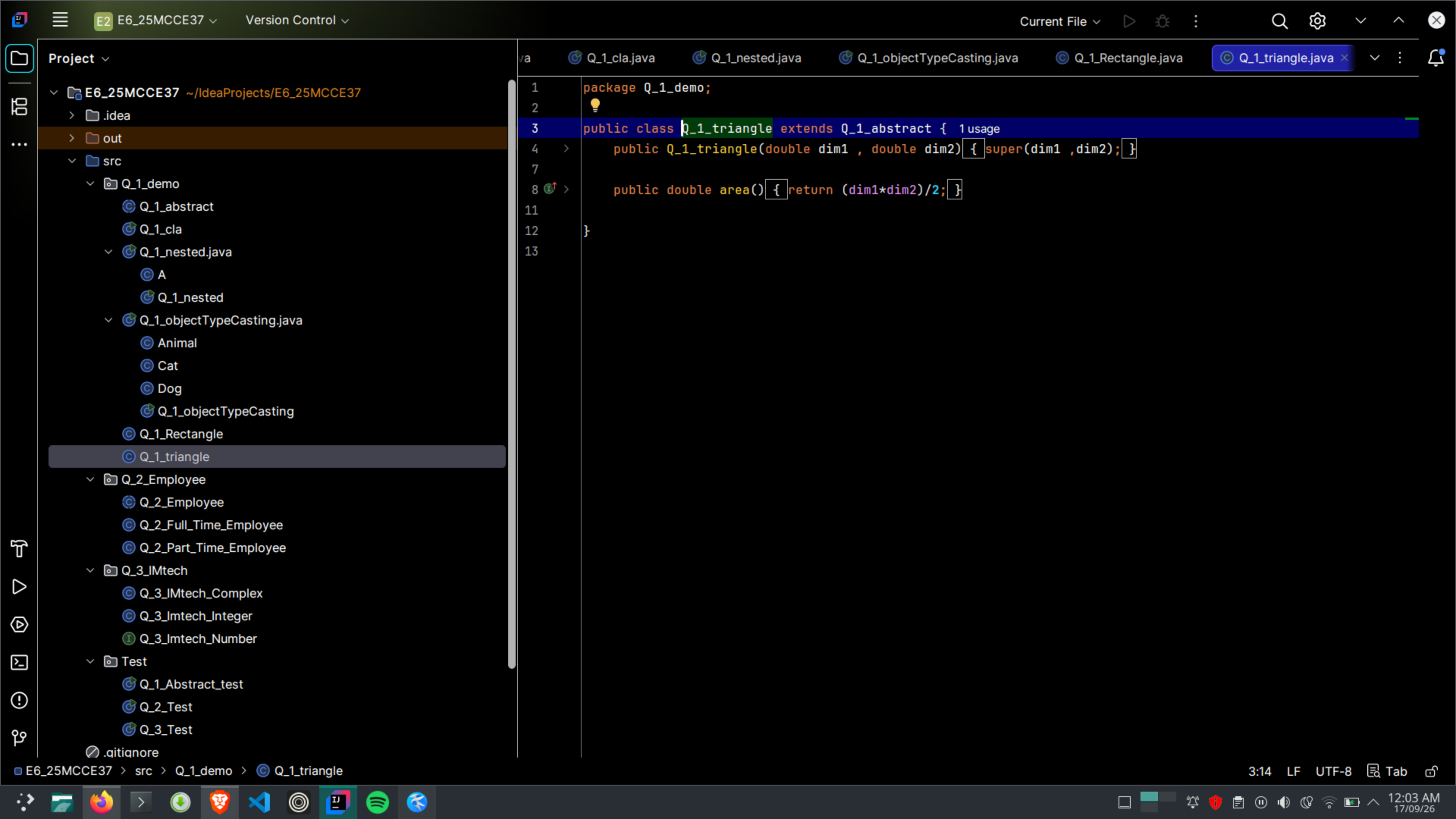
The image shows a screenshot of an IDE, likely IntelliJ IDEA, with a dark theme. The top bar displays the project name 'E6_25MCCE37' and the 'Version Control' tab. The left sidebar shows the project structure, with the 'src' directory expanded to show the 'Q_1_demo' package. Inside 'Q_1_demo', the file 'Q_1_nested.java' is selected, and the class 'A' is highlighted. The main editor window shows the code for 'Q_1_nested.java'. The code defines a package 'Q_1_demo', a class 'A' with an integer 'a' and a 'display()' method, and a nested class 'B' with a 'displayB()' method. The 'display()' method in 'A' calls 'displayB()' on a new instance of 'B'. The 'displayB()' method in 'B' calls 'display()' on 'this' and prints a message. The 'main()' method in 'Q_1_nested' creates an instance of 'A', creates an instance of 'B' inside 'A', and calls 'displayB()' on it. The bottom status bar shows the file path 'E6_25MCCE37 > src > Q_1_demo > Q_1_nested.java > A' and the encoding 'UTF-8'.

The screenshot shows an IDE with the following components:

- Top Bar:** Displays the current file 'Q_1_nested.java' and other open files like 'Q_1_cla.java', 'Q_1_abstract.java', 'Q_3_Test.java', and 'Q_3_IMtech_Complex.java'. It also shows a 'Current File' dropdown and a 'Version Control' dropdown.
- Left Panel (Project View):** Shows the project structure for 'E6_25MCCE37'. The 'src' directory is expanded, showing the 'Q_1_demo' package. Inside 'Q_1_demo', the file 'Q_1_nested.java' is selected, and the class 'A' is highlighted.
- Main Editor:** Displays the code for 'Q_1_nested.java'. The code defines a class 'A' with an inner class 'B'. The 'B' class has a 'displayB()' method that prints 'In the Inner class a = ' followed by the value of 'a'. The 'A' class has a 'main()' method that creates an instance of 'A', creates an instance of 'B' (labeled 'b2'), and calls 'displayB()' on it. The 'Q_1_nested' class is a public class with a 'main()' method that creates an instance of 'A', creates an instance of 'B' (labeled 'b2'), and calls 'displayB()' on it.
- Bottom Panel:** Shows the breadcrumb 'E6_25MCCE37 > src > Q_1_demo > Q_1_nested.java > A' and the status bar with '4:7 LF UTF-8 Tab'.







New Tab Split View

Copy Paste Find

```
> ls
Desktop  Downloads  hp-check.log  Music      package.json  Pictures  Public      strap.sh  unnecessary  Videos
Documents  foo.txt    IdeaProjects  node_modules  package-lock.json  Projects  Screenshot_20260906_003007-1.png  Templates  Untitled

> cd Desktop
> cd sem3
> cd oop
> ls
books_references  lab_assignments
> cd la
cd: no such file or directory: la
> cd lab_assignments
> ls
25mcce37_E2  25mcce37_E2.zip  25mcce37_EX1  25mcce37_EX1.zip  25MCCE37_EX3  25MCCE37_EX3.zip  E4_25MCCE37  E4_25MCCE37.zip  E5_25MCCE37  E5_25MCCE37.zip  E6_25MCCE37
> cd E6_25MCCE37
> ls
1.png  2.png  3.png  4.png  5.png  6.png  7.png  E6_25MCCE37
> cd ..
> cd /
> ls
bin  boot  desktopfs-pkgs.txt  dev  etc  home  lib  lib64  mnt  opt  proc  root  rootfs-pkgs.txt  run  sbin  srv  sys  tmp  usr  var
> cd home
> cd Desktop
cd: no such file or directory: Desktop
> cd vaibhav
> ls
Desktop  Downloads  hp-check.log  Music      package.json  Pictures  Public      strap.sh  unnecessary  Videos
Documents  foo.txt    IdeaProjects  node_modules  package-lock.json  Projects  Screenshot_20260906_003007-1.png  Templates  Untitled
> cd IdeaProjects
> ls
24MCCE49_EX3  25mcce37_E2  25MCCE37_E4  25mcce37_EX1  25MCCE37_EX3  E5_25MCCE37  E6_25MCCE37  java_online_course
> cd E6_25MCCE37
> ls
E6_25MCCE37.iml  out  src
> cd src
> ls
Q_1_demo  Q_2_Employee  Q_3_IMtech  Test
> cd Q_1_demo
> javac Q_1_cla.java
> java Q_1_cla.java vaibhav apple cat orange
Total arguments: 4
Argument 0:vaibhav
Argument 1:apple
Argument 2:cat
Argument 3:orange
```

~/IdeaProjects/E6_25MCCE37/src/Q_1_demo



The image shows a screenshot of an IDE interface. On the left is a project explorer with a tree view of files and folders. The 'Test' folder is expanded, and 'Q_1_Abstract_test' is selected. The main editor area shows the code for 'Q_1_Abstract_test.java'. The code includes a package declaration 'package Test;', imports for 'Q_1_Abstract' and 'Q_1_Rectangle', and a 'main' method that creates instances of these classes and prints the area of a rectangle. The bottom status bar shows the current file path 'E6_25MCCE37 > src > Test > Q_1_Abstract_test' and the time '6:14'.

The screenshot displays an IDE interface with a project explorer on the left and a code editor on the right. The project explorer shows a project named 'Project' with the following structure:

- Q_1_abstract
- Q_1_cla
- Q_1_cla.class
- Q_1_nested.java
 - A
 - Q_1_nested
- Q_1_objectTypeCasting.java
 - Animal
 - Cat
 - Dog
 - Q_1_objectTypeCasting
- Q_1_Rectangle
- Q_1_triangle

The code editor shows the file 'Q_1_Abstract_test.java' with the following code:

```
1 package Test;
2 import ...
5
6 public class Q_1_Abstract_test {
7     public static void main(String[] args) {
8         // Q_1_abstract a1 = new Q_1_abstract(1,3);
9
10        Q_1_Rectangle r1 = new Q_1_Rectangle(1,3);
11
12        System.out.println("area of the rectangle is " + r1.area());
13    }
14 }
15
```

Run

Q_1_Abstract_test



```
/usr/lib/jvm/default/bin/java -javaagent:/usr/share/idea/lib/idea_rt.jar=42259 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath /home/va
area of the rectangle is 3.0
```

```
Process finished with exit code 0
```

1

